Contamination Model: Team Update

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Our prototype now runs end to end: on simulated batches it flagged every contaminated batch, typically about 7.5 hours before the visual/odour check would have, explained the likely cause, and pointed to the shared medium lot behind most cases. It has not yet been tested on real MaverickX data.

In plain words

The system is a second pair of eyes on every enzyme batch. Today MaverickX spots contamination by smell, colour and experience, often after the batch is already lost. Our system watches the routine measurements (pH, temperature, dissolved oxygen, biomass and enzyme activity) and answers four questions for the scientist:

1. **Is this batch behaving unusually?** It compares the batch with what normal batches looked like at the same hour.
2. **What is the most likely reason?** Contamination, slow but normal growth, a change in raw materials or settings, or a faulty sensor.
3. **What should we check next?** It suggests the quickest, cheapest test that tells the likely reasons apart.
4. **Where might it have come from?** It looks for a medium lot, inoculum lot or bioreactor that the contaminated batches have in common.

It supports the scientist's decision; it never discards a batch or declares contamination on its own. Only a real lab test can mark contamination as confirmed.

How it works

Each batch goes through six steps, from raw readings to a one-page investigation report.

1. **Check the data.** Missing, impossible, duplicated or wrong-unit readings are flagged, never deleted or filled with zero, so no evidence is lost silently.
2. **Learn what normal looks like.** From 30 batches the scientists approve as normal, it learns the typical value of each measurement at each hour and how much it usually varies. This reference is then frozen and versioned, so slow drift can't quietly become the new normal.
3. **Spot unusual behaviour.** Every new reading is compared with the reference. Three simple, explainable detectors raise alerts: one reading far from normal, a sustained drift, and an accumulating shift. Each alert records what changed, by how much, when,

and which other signals moved with it.

4. **Suggest explanations.** A rule book that scientists can read and edit links patterns of evidence to explanations. For example, extra biomass plus extra oxygen use plus a more acidic culture points to contamination; one signal drifting alone points to a faulty probe. Explanations get a transparent score, not a probability. When the evidence is thin, the honest answer is "unresolved, too early to tell".
5. **Recommend the next check.** It picks the check that best separates the leading explanations (e.g. microscopy versus a probe check), and keeps a contamination check on the list until contamination is ruled out.
6. **Trace the source.** It compares how often contamination happened in batches that used each medium lot, inoculum lot and bioreactor versus batches that did not. A resource that stands out is reported as a lead to investigate, not proof.

When a scientist records a test result (say, microscopy finds foreign cells), the assessment updates straight away.

Results so far

On 20 fresh simulated campaigns (500 test batches) that were not used to tune anything, the system caught every contaminated batch and named the right cause for 296 of the 300 flagged batches by the end of the batch.

What we measured	Result
Contaminated batches caught	120 of 120
Warning before the visual/odour check	about 7.5 h earlier (median; 5–9 h across campaigns)
Sensor faults caught (spike or drifting probe)	120 of 120
Normal-category batches flagged	81 of 260 (60 of them are genuinely slow-growing batches)
Right explanation, 8 h after the first alert	241 of 300 (6 wrong, the rest "too early to tell")
Right explanation, end of batch	296 of 300 (3 unresolved, 1 wrong)
Shared medium lot found as the top lead	20 of 20 campaigns
Campaigns with a false strong lead	1 of 20 (a real chance cluster)

At the very first alert the system mostly says "too early to tell", on purpose: one signal moving alone fits contamination, a faulty probe and normal variation alike. Full numbers and method are in `docs/mvp-results.md` in the repo.

The dashboard

Scientists use a web dashboard with four pages; no coding needed.

Page	What a scientist can do
Campaign overview	See every batch at a glance: which are flagged, the likely cause, whether contamination is confirmed, and which other batches are linked
Batch investigation	See each measurement against the normal band with alerts marked; replay the batch hour by hour to see what the system knew at that time; read the ranked explanations, evidence and next checks; record a test result and watch the assessment update; download the report
Source tracing	See which lots or bioreactors the contaminated batches share, drawn as a graph, with the statistics behind each lead
How it works	A plain-language explanation of the method and its limits

Limits

The results show the software works as designed. They do not yet show it works on MaverickX's real process.

- **Simulated data only.** We used α-amylase batches with E. coli as the example contaminant, because we have no MaverickX batch records yet. The contaminant's effects (extra biomass, extra oxygen use, acidity, lower activity) are our assumptions.
- **The rule book is a draft.** Every rule, weight and suggested check was written by us and needs review by MaverickX scientists. Check times and costs are placeholders.
- **It flags unusual batches, not organisms.** It never identifies a species from pH or oxygen readings, and its scores are not probabilities.

- **False alarms remain.** Legitimately slow batches get flagged too, and there are about 2 alerts per batch that don't point to real contamination. The explanations step is what keeps these from being mistaken for contamination.
- **Source tracing sees only what is recorded:** medium lot, inoculum lot and bioreactor. Water, transfer lines and environmental swabs aren't included, and with small campaigns a chance cluster can look like a real source.
- **It investigates; it doesn't predict.** Predicting contamination before it starts would need historical data with confirmed cases.

What we need from the team

The prototype is ready to show; the deliverables are due `Oct 28, 2026`.

- [ ] Review the rule book (`src/contamination/knowledge/catalogue.yaml`): are the contamination signs and weights biologically sensible? (Aisha, Beren)
- [ ] Confirm the proxy choice: α-amylase with E. coli, or lipase with sulfate-reducing bacteria
- [ ] Send MaverickX a data request: past batch records, measurements, timestamps, stages, lots/equipment used, and how contamination was confirmed
- [ ] Ask MaverickX which tests they can actually run, with real turnaround times and costs, to replace our placeholders
- [ ] Use the dashboard and the example report in the one-pager and technical document
- [ ] Share with Pranav and Shea and the validators for feedback

How to run it

Everything is in the triple-ignition-challenge repo and needs Python 3.11 or newer.

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.lock && pip install -e ".[dev]"
python -m contamination.demo      # generates the simulated batches and
reports
streamlit run app/dashboard.py    # opens the dashboard at
http://localhost:8501
```

The README has the code map, and `docs/example-investigation-report.md` shows a full report for one contaminated batch.